	lang	capi	cnty	col	num	mon
G-A	7.46	6.61	6.62	4.10	2.75	4.82
G-F	0.63	1.89	1.67	3.51	4.70	2.81
L-A	6.28	5.03	6.42	3.81	2.22	5.10
L-F	1.74	4.30	2.90	4.02	5.60	3.05

Table 3: Contribution of attention and FFN layers.

layer for each knowledge in Table 4, where a_l and f_l means lth attention and FFN layer. For a clearer display, we illustrate the importance (darker color means larger importance) of top10 layers in Figure 4 (GPT2) and Figure 5 (Llama).

Both attention and FFN layers have ability to store knowledge, and all the top10 layers are in deep layers. Information with analogous semantics (e.g., language, capital, country) tends to be stored within similar layers/modules. For instance, a_{26}, a_{30}, a_{28}, and a_{22} in GPT2 ranks top for language, capital and country, and a_{23} in Llama-7B ranks the first for these knowledge. Data with dissimilar semantics (e.g., language, color, month) typically resides in distinct layers/modules.

	top10 important layers
lang	$a_{26}, a_{30}, a_{32}, a_{22}, a_{31}, a_{28}, a_{23}, a_{27}, a_{19}, a_{23}$
capi	$a_{26}, a_{28}, a_{30}, a_{25}, a_{22}, f_{26}, f_{28}, a_{19}, f_{27}, f_{30}$
cnty	$a_{26}, a_{30}, a_{28}, a_{22}, f_{29}, a_{31}, f_{26}, a_{32}, a_{25}, a_{19}$
col	$a_{32}, f_{32}, a_{33}, f_{29}, f_{31}, a_{31}, a_{26}, f_{33}, f_{28}, a_{22}$
num	$f_{29}, f_{23}, f_{27}, f_{30}, f_{31}, f_{26}, f_{32}, a_{23}, a_{22}, f_{28}$
mon	$a_{27}, a_{26}, f_{26}, a_{25}, f_{30}, a_{28}, a_{24}, a_{22}, a_{30}, f_{27}$
lang	$a_{23}, a_{21}, f_{21}, a_{19}, a_{18}, a_{31}, a_{25}, a_{16}, f_{20}, f_{19}$
capi	$a_{23}, f_{21}, f_{22}, a_{18}, a_{25}, a_{21}, f_{19}, f_{20}, a_{16}, f_{24}$
cnty	$a_{23}, a_{21}, a_{25}, f_{22}, a_{18}, a_{19}, a_{16}, f_{21}, f_{31}, a_{31}$
col	$f_{29}, a_{20}, f_{22}, f_{20}, a_{19}, a_{28}, a_{16}, a_{29}, a_{18}, f_{28}$
num	$f_{31}, f_{26}, f_{29}, f_{27}, a_{26}, f_{23}, f_{24}, a_{28}, f_{17}, f_{30}$
mon	$a_{21}, a_{19}, f_{19}, a_{16}, f_{31}, a_{23}, a_{28}, f_{30}, f_{17}, f_{18}$

Table 4: Top10 important layers in GPT2 (first block) and Llama (second block).

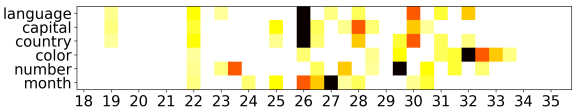

Figure 4: Top10 important "value layers" in GPT2.

Head-level knowledge storage. We compute the importance score of each head (detailed in Appendix B) and find that many heads have ability to store similar knowledge. In GPT2, a^6_{30} (30th layer 6th head), $a^{17}_{26}, a^{11}_{32}, a^{13}_{25}$ and a^{17}_{22} rank top8 for language, capital and country. Similarly, $a^{12}_{23}, a^{31}_{19}, a^{25}_{31}$, and a^{25}_{25} rank top5 for these knowledge in Llama.

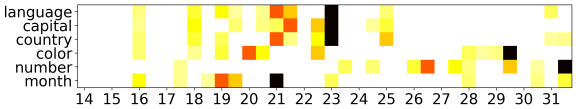

Figure 5: Top10 important "value layers" in Llama.

To evaluate how much knowledge the top heads store, we intervene the top 1% heads (top7 in GPT2 and top10 in Llama) by setting the heads' parameters to zero. Intervening each knowledge's heads result in a MRR/probability decrease of 44.5%/53.3% in GPT2, and 32.8%/48.2% in Llama (shown in Appendix B). But semantic-unrelated knowledge only reduce 7.1%/9.5% in GPT2 and 3.8%/8.7% in Llama. Therefore, the identified "knowledge heads" contain much semantic-related knowledge.

	lang	capi	cnty	col	num	mon
(attn) all	6.7	5.5	6.9	4.0	2.4	5.4
positive	30.5	29.4	30.5	32.0	24.8	29.2
top100	3.5	3.0	3.6	2.8	2.0	2.8
top200	5.0	4.4	5.0	4.1	3.0	4.1
(FFN) all	2.5	5.1	3.6	4.9	6.5	2.8
positive	77.4	77.6	69.8	90.6	71.6	69.1
top100	6.4	6.3	5.5	6.1	6.6	7.0
top200	8.2	8.0	7.0	8.0	8.5	8.5

Table 5: Importance of top neurons in attention (first block) and FFN (second block) layers in Llama-7B.

Neuron-level knowledge storage. For attention and FFN layers in Llama, we compute the sum of importance score for all neurons, all positive neurons (score larger than 0), top100 neurons, and top200 neurons, as illustrated in Table 5. Similar results of GPT2 is shown in Appendix C.

In both models, the sum score of top200 neurons in attention layers and top100 neurons in FFN layers are similar to that of all neurons. Additionally, we intervene the top neurons to evaluate how much final predictions are affected, detailed in Appendix C. When intervening the top200 attention neurons and top100 FFN neurons for each sentence, the MRR and probability decreases 96.3%/99.2% in GPT2, and 96.9%/99.6% in Llama. In comparison, randomly intervening the same number of neurons only decreases 0.22%/0.14%. Hence, even though there are many neurons contribute to the final prediction, intervening a few neurons (300) affects the final prediction much. This conclusion holds significance for future studies delving into neuron-level knowledge editing.

	top10 query layers for FFN neurons
lang	$a_{26}, a_{22}, a_{19}, f_{26}, a_{17}, f_{25}, f_{27}, f_{23}, a_{25}, a_{23}$
capi	$a_{26}, a_{22}, a_{19}, a_{17}, f_{25}, a_{23}, a_{18}, f_{26}, f_{21}, a_{28}$
cnty	$a_{26}, a_{22}, a_{17}, a_{19}, a_{23}, f_{18}, a_{20}, a_{18}, f_{21}, f_{25}$
col	$f_{26}, f_{29}, a_{26}, f_{28}, f_{25}, a_{22}, f_{27}, a_{17}, a_{24}, f_{23}$
num	$f_{26}, f_{23}, f_{27}, a_{22}, f_{25}, a_{17}, f_{19}, f_{21}, a_{23}, f_{0}$
mon	$a_{17}, a_{22}, f_{23}, a_{26}, f_{26}, f_{24}, a_{19}, a_{20}, f_{20}, f_{21}$
lang	$f_{21}, a_{16}, a_{19}, f_{18}, a_{18}, a_{21}, a_{17}, f_{30}, f_{19}, a_{14}$
capi	$a_{18}, a_{16}, f_{23}, a_{19}, f_{17}, a_{14}, f_{22}, a_{21}, f_{26}, f_{19}$
cnty	$a_{16}, a_{18}, a_{21}, a_{19}, f_{21}, a_{14}, f_{19}, a_{17}, a_{31}, f_{20}$
col	$f_{20}, f_{21}, a_{15}, a_{17}, a_{18}, a_{20}, f_{19}, f_{22}, f_{17}, a_{16}$
num	$f_{19}, f_{21}, f_{22}, f_{16}, a_{18}, a_{22}, a_{24}, f_{14}, a_{12}, a_{25}$
mon	$a_{16}, a_{19}, f_{18}, a_{21}, a_{17}, a_{14}, f_{29}, f_{19}, f_{17}, a_{18}$

Table 6: Top10 query layers for top100 FFN neurons in GPT2 (first block) and Llama (second block).

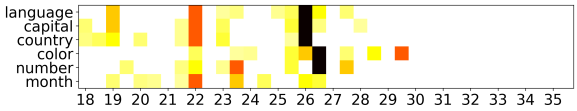

Figure 6: Top10 important "query layers" in GPT2.

Important query layers for FFN value neurons. The "value" FFN neurons are activated by last position's residual stream. We evaluate which "query layers" activate the top100 FFN neurons, shown in Table 6. We also illustrate the importance of top10 important "query layers" in Figure 6 (GPT2) and Figure 7 (Llama). The medium-deep attention layers play large rules. Compared Figure 6-7 with Figure 4-5, we find that several attention "query layers" also contribute to final predictions (e.g. a_{19}, a_{22}, a_{26} in GPT2 and $a_{16}, a_{18}, a_{19}, a_{21}$ in Llama for country/capital/language). These medium-deep attention layers' neurons are very important, working as both "value" and "query".

Important query neurons for attention value neurons. We compute the important query layers activating the top200 "value" attention neurons. finding the shallow and medium FFN layers play main roles (detailed in Appendix D). To identify the important query FFN neurons, we weighted sum the inner product between each attention neuron's subkey and each FFN neuron on every position's residual stream, as query FFN neurons' scores. When intervening top1000 shallow neurons

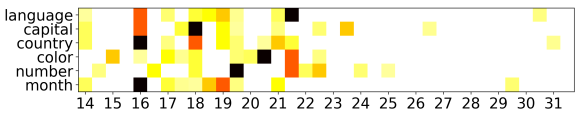

Figure 7: Top10 important "query layers" in Llama.

	lang	capi	cnty	col	num	mon
G	91/96	98/99	89/94	95/97	96/96	83/88
L	78/92	84/93	91/98	85/93	90/96	94/98

Table 7: MRR/probability decrease (%) when intervening 1,000 query neurons in GPT2 (G) and Llama (L).

for each sentence, both MRR and probability drops very much (92%/95% in GPT2 and 87%/95% in Llama), shown in Table 7. In comparison, randomly intervening 1,000 neurons only result in a decrease of 0.8%/1.1%. Hence, our method can locate the important query neurons in these layers.

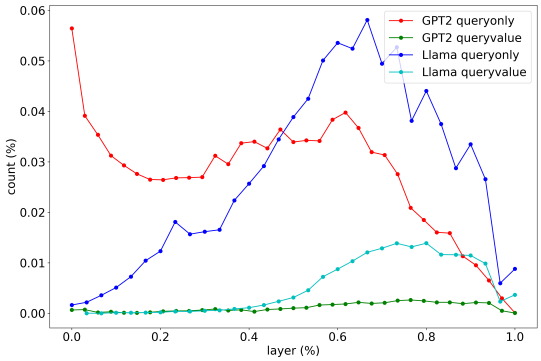

Figure 8: Query neuron distribution in GPT2 and Llama.

Then we count the number of queryvalue (both in top1000 query and top1000 value) and queryonly (only in top1000 query) FFN neurons, shown in Figure 8. In both models, the number of queryonly neurons, which is much larger than that of queryvalue neurons, starts to drop at 60% layer. This observation indicates that the shallow and medium FFN neurons are important for activating the attention "value neurons". A difference is that the very shallow FFN layers play large roles in GPT2, and we defer this exploration to future research. Overall, our analysis learns the information flow at neuron level: features in shallow/medium FFN neurons are extracted, then activate the deep attention and FFN neurons related to final predictions.

	lang	capi	cnty	col	num	mon
G-query	5	51	26	17	104	81
G-value	48	71	64	57	121	137
L-query	1	1	1	9	39	44
L-value	13	18	21	23	84	95

Table 8: Shared neurons in GPT2 (G) and Llama (L).

Shared Value and query neurons in each knowledge. We compute how many "shared" query neurons and value neurons rank top300 in more than

50% sentences in each knowledge, shown in Table 8. On average, there are 27.6% shared value neurons in GPT2 and 14.1% in Llama. Query neurons, with 15.7% shared neurons in GPT2 and 5.2% in Llama, exhibit a more dispersed distribution than value neurons. To explore the neurons' interpretability, we project them into vocabulary space. We find most value neurons (first block in Table 9) are related to predicted tokens. However, we do not observe much interpretability in query neurons. We only find a few query neurons (second block in Table 9) related to the final words. Hence, to explore the interpretability of query neurons may be a valuable direction in future works.

neuron	top10 tokens in vocabulary space
f_{29}-3771 (GPT2,v)	Chile,Nicaragua,Finland,Ireland,Belarus, Norway,Slovakia,Latvia,Australia
a_{23}^{12}-70 (Llama,v)	German,Greek,Netherlands,Dutch, Germany,Greece,Holland,Norwegian
f_0-2947 (GPT2,q)	Lion, Bull, Liver, riot, Gladiator, **Red**, uct, Ant, Les
f_7-8744 (Llama,q)	**Belgium**,Ireland,Vienna,Czech,Kas, Kansas,Netherlands,Iowa,wings,Spanish

Table 9: Interpretable neurons in vocabulary space.

5 Conclusion

In this study, we propose a method based on log probability increase to identify the important "value neurons". We also develop a method based on inner products to locate the "query neurons" activating these "value neurons". Our method and analysis on six types of knowledge are helpful for exploring and understanding the mechanism of LLMs.

6 Limitations

The first limitation of our study is that it focuses on six specific types of knowledge, while other types of knowledge are also important. Secondly, our experiments are conducted using GPT2-large and Llama-7B models. It is essential to compare the similarities and differences in knowledge storage across different models. Lastly, our study employs static methods for neuron-level knowledge attribution. Although our experiments demonstrate the correctness and robustness of our designed method, it is also important to compare static methods with other attribution methods, such as causal mediation analysis and gradient-based methods. We plan to explore these areas in future work.

A potential risk of our work is that people can utilize our method to identify important neurons and edit them to change the models' outputs. For instance, if they identify the toxicity neurons and gender bias neurons and increase these neurons' coefficient scores, the model will be more likely to generate toxicity and gender bias words. But this potential risk depends on how people utilize our method. Our method can be utilized for reducing hallucinations, toxicity, and bias in LLMs by identifying and intervening/editing these neurons.

7 Acknowledgements

We thank Kailai Yang, Zhiwei Liu, and John McNaught for helpful feedbacks and constructive suggestions. This work is supported by the project JPNP20006 from New Energy and Industrial Technology Development Organization (NEDO). This work is supported by the computational shared facility and the studentship from the Department of Computer Science at the University of Manchester.

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